Implicit degree bias in the link prediction task

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Summary

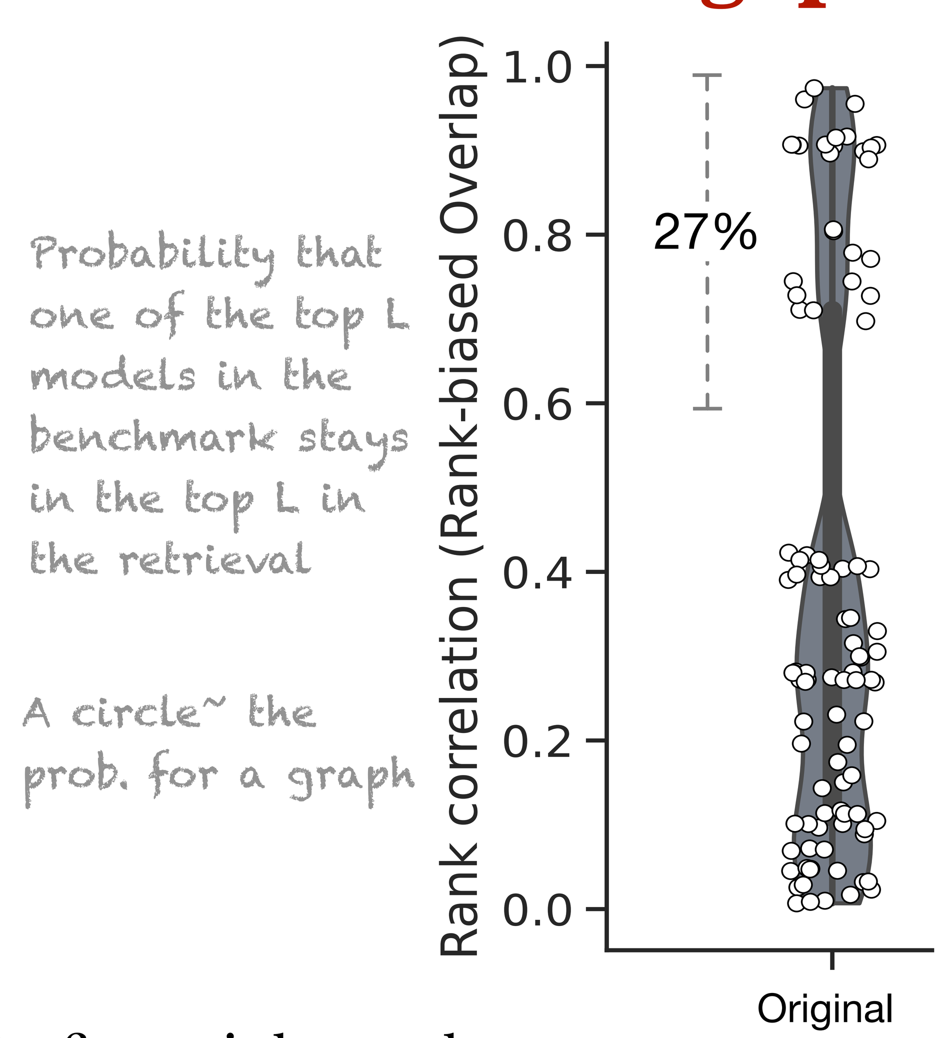
- **Problem:** Standard link prediction benchmarks suffer from implicit degree bias, allowing models to achieve high performance by simply predicting edges between high-degree nodes rather than learning meaningful structural patterns.
- **Evidence:** Evaluation of 27 models across 95 graphs reveals this fundamental flaw in current evaluation practices.
- **Impact**: Benchmarks favor trivial degree-based predictions over structural learning, misleading model comparison and development.

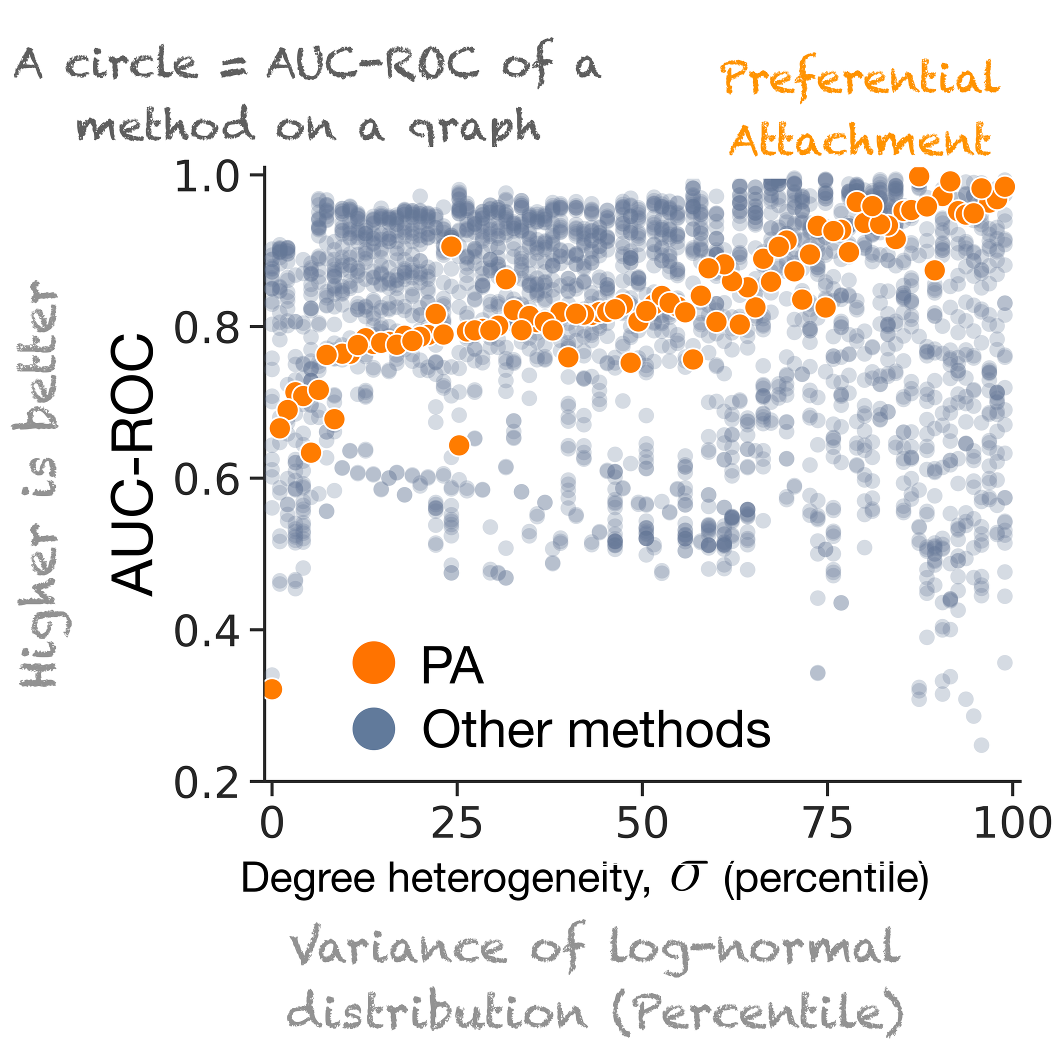
Benchmark vs. Real-world Performance gap

Evaluated 27 models across 95 networks on both standard benchmarks and practical retrieval tasks (top-K recommendations, like friend suggestions).

Key Finding: 60% of top models in the benchmark are not the top retriever across 70% of graphs..

Implication: Standard benchmark scores poorly predict real-world effectiveness.





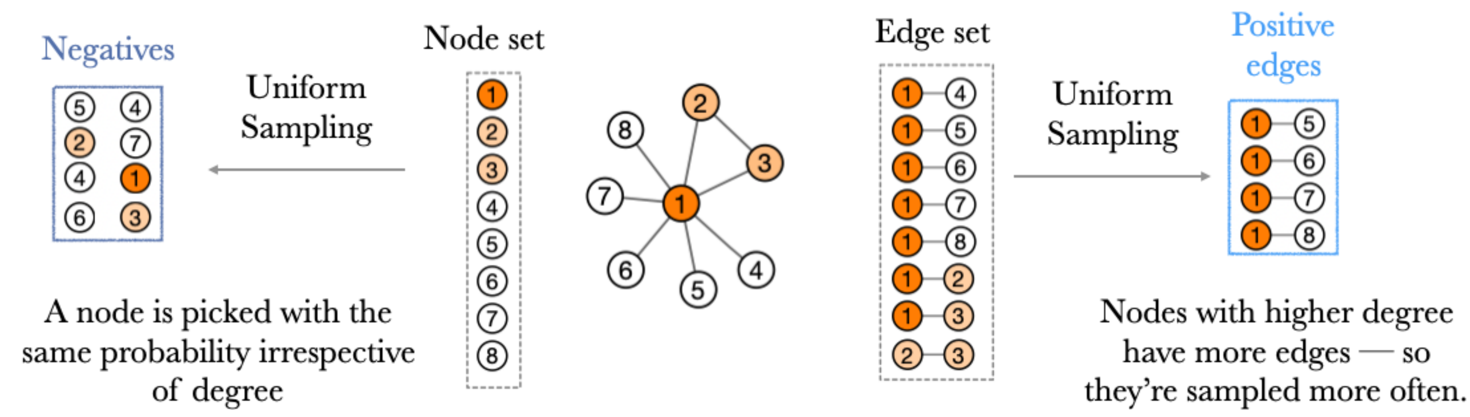
Preferential attachment (score=$k_i k_j$) predicts edges using *only* node degrees yet achieves exceptionally good benchmark performance:

- Scisci Citation: 0.94
- USPTO Citation: 0.91
- OGBL Protein-Protein: 0.93
- OBGL Drug-Drug: 0.99 🤯

Nearly maximum performance especially for degree-hetero. graph.

Degree-bias in the benchmark

Sampling Asymmetry: In the benchmark, methods are tested on the positive and negative edges sampled from *different sets.*



Result: High-degree nodes appear more frequently in the positives (since they have more edges) but equally with small degree nodes in the negatives.

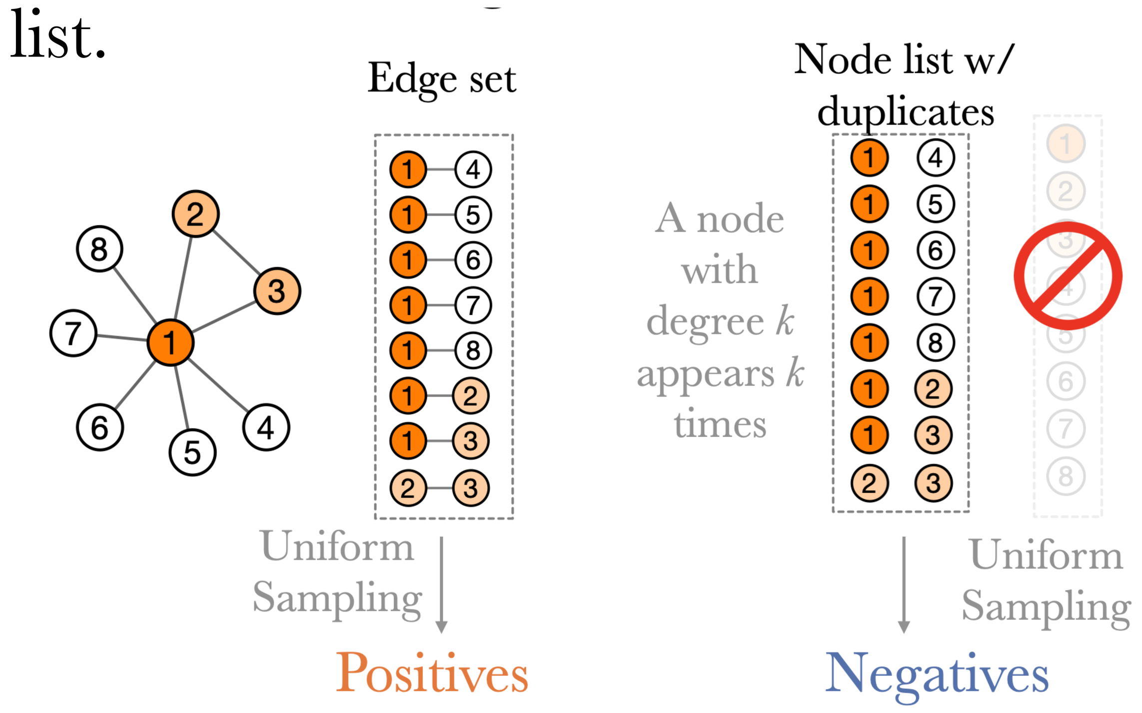
Consequence: Degree-based heuristics trivially distinguish positive/negative samples by node degree alone, inflating benchmark performance.

Solution

Current benchmarks compare positive edges to random node pairs - like comparing disease patients to random people instead of matched controls.

Our Method: Sample negative edges with same degree distribution as positive edges.

Implementation: Create node list with k copies of degree-k nodes → sample negative edges from this list.



Result: Eliminates degree bias, enabling fair model comparison.

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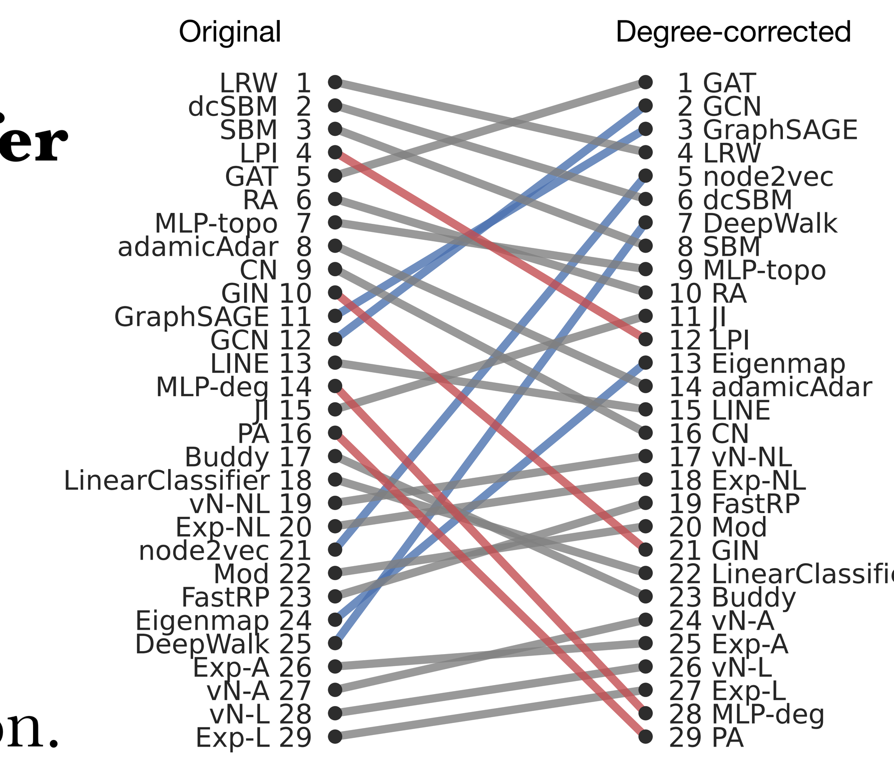
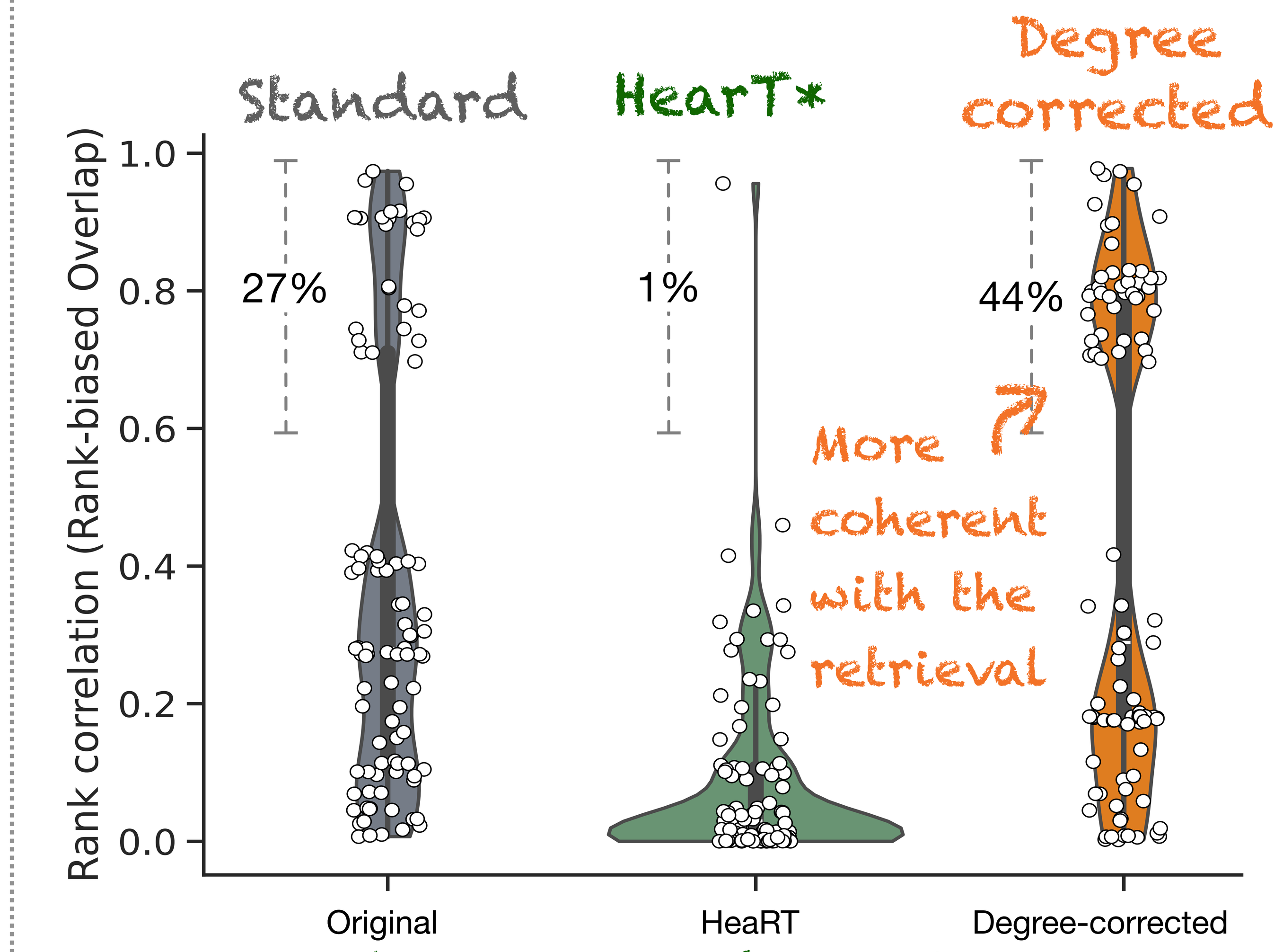
More 🚀🚀: Find our paper on the mathematical analysis, validation analysis, and effectiveness of degree-correction for community detection!

Results

Rankings differ (average AUR-ROC): PA, LPI, and GIN drop rankings while GCN, node2vec improved by degree-correction.



Better alignment: The degree-corrected benchmark improves the alignment between benchmarks and retrieval tasks (more networks with high RBO scores).



* Correct the distributional mismatches of distances (Li et al. NeurIPS 2023)

Biased sampling restores uniformity: We divide nodes into two groups by degree and count # of edges between and within the groups.

>70% of AUC-ROC is attributed to trivial high vs. low-degree comparisons in the standard benchmark for degree hetero. graph. Degree-correction <40% for most graphs.